SMS Spam Detection using Machine Learning

Abstract

Spam messages are a major issue in digital communication, causing inconvenience and potential security risks. This project focuses on developing a machine learning model to classify SMS messages as spam or ham (legitimate). Natural Language Processing techniques were applied to preprocess text data, followed by feature extraction using TF-IDF vectorization. A Multinomial Naive Bayes classifier was trained and achieved high accuracy in prediction.

Introduction

With the rapid growth of mobile communication, spam messages have become increasingly common. These unwanted messages often contain advertisements or malicious content. Manual filtering is inefficient, creating the need for automated spam detection systems. Machine learning provides intelligent methods to analyze patterns in text data and classify messages accurately.

Objectives

- To preprocess SMS text data using NLP techniques
- To convert text into numerical features
- To train a machine learning model for classification
- To evaluate model performance using metrics
- To detect whether a message is spam or ham

Dataset Description

The dataset used is the SMS Spam Collection Dataset, containing labeled SMS messages categorized as spam or ham. Each record includes a label and the corresponding message text. The dataset was obtained from Kaggle.

Methodology

- Data Loading using Pandas
- Text preprocessing: lowercase conversion, punctuation removal, stopword removal
- Feature extraction using TF-IDF vectorization
- Model training using Multinomial Naive Bayes
- Model evaluation using accuracy, precision, recall, F1-score, and confusion matrix

Tools and Technologies Used

- Python
- Pandas and NumPy
- Scikit-learn
- NLTK
- Matplotlib
- Machine Learning
- Natural Language Processing

Results

The trained model achieved approximately 97% accuracy, demonstrating strong performance in detecting spam messages. The evaluation metrics confirm the effectiveness of the model.

Conclusion

This project demonstrates how machine learning and NLP techniques can be applied to automatically detect spam messages. The system achieved high accuracy and can be extended for real-time applications.

Future Scope

- Integration with mobile or web applications
- Real-time spam detection system
- Use of deep learning models such as LSTM
- Deployment using web frameworks

References

- Kaggle Dataset
- Scikit-learn Documentation
- NLTK Documentation
- Python Libraries Documentation